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Palaeomagnetic dating of haematite genesis in Orcadian Basin sediments

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Synopsis

Stable directions of remanence in samples of the red facies of the Old Red Sandstone in the Caithness region correspond to a complex magnetisation mainly acquired in Carbo-Permian times. This indicates that haematite in these sediments formed or passed through a growth stage in late Palaeozoic Permian times that obscured most pre-existing magnetisation and was not seriously affected by any later chemical changes. The timing of this period of haematite growth illustrates the application of palaeomagnetic techniques to the dating of certain post depositional processes in sedimentary rocks and raises problems about the mechanism involved in such changes in the case of the Old Red Sandstones of northern Scotland.

Introduction

Sediments may acquire a remanent magnetisation in a variety of ways and each process causes a magnetic alignment with the geomagnetic field at that time which, under favourable circumstances, may be preserved to the present day (Tarling, 1971). In particular, haematite grains, a few Angstrom's in diameter, acquire a very stable magnetisation as they grow to sizes of the order of 10^{-5} to 10^{-3} cm diameter. This means that it is possible to date the age of certain processes by comparing the directions remanent magnetisation with known polar-wandering paths or with polarity reversal sequences. In particular, those diagenetic processes that result in magnetisation can be dated. In this study, the Devonian strata from northern Britain were originally investigated in order to define the direction of the geomagnetic field in Britain during Devonian times, but it became clear that most of the stable magnetisation was carried by haematite that must have formed in Permo-Carboniferous times, rather than Devonian. This observation did not, at first sight, appear consistent with sedimentary and palaeoenvironmental features which suggested that the original environment of deposition had been oxidising and that the haematite was of Devonian age. The results presented here outline the apparently contradictory evidence in this region as an illustration of the application of palaeomagnetic dating techniques to problems in diagenesis and also initiates a more detailed study of chemical processes affecting haematite growth in the Devonian sediments of northern Britain.

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Scott. J. Geol. 12, (2), 125–134, 1976

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Palaeomagnetic data

Hand samples and drilled cores were oriented in the field at a variety of sites (Fig. 1) from Dunnet Head to south of the Moray Firth. The samples were cut into 2·5 cm diameter, 2·5 cm high cylindrical specimens and their initial magnetisation was measured using astatic and spinner magnetometers. One representative pilot specimen was chosen from each site and subjected to routine incremental thermal demagnetisation (Tarling, 1971) up to 650°C in order to determine their blocking temperature spectra. Thermally induced chemical changes of the magnetic minerals during heating were monitored by measuring the susceptibility (Collinson *et al.* 1963) after each temperature increment. On the basis of the behaviour of the pilot specimens, the remaining specimens were all heated to 300°C and their magnetisation remeasured. As titanomagnetites have Curie points lower than 600°C, most of the collection was then heated to between 600 and 650°C, and remeasured, in order to examine the high temperature remanence carried by haematite alone.

(i) *Dunnet Head Sandstone* comprises 600 m of pink and yellow unfossiliferous, moderately consolidated sandstones that dip 12° northwards. The age of these rocks has recently been determined as Upper Devonian (McAlpine, pers. comm.) The succession was sampled at 18 sites (1–18, Fig. 1) each being at a separate stratigraphical level and comprising six separately oriented samples. The magnetic intensity and susceptibility of the specimens were low, 10^{-7} gauss and 10^{-7} gauss oersted^{-1}, and the

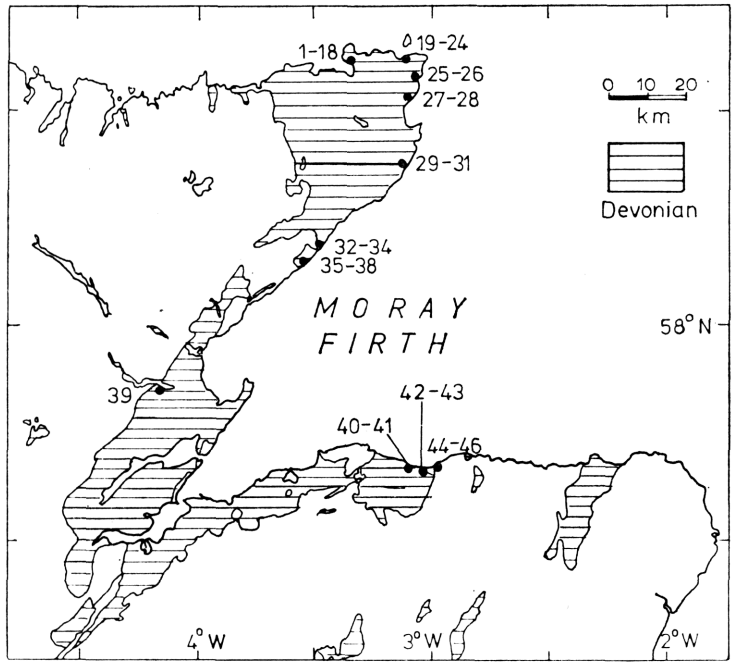


Fig. 1. Site locations in northern Scotland. The areas of Devonian rocks are shown shaded.

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initial directions were very scattered. Pilot specimens did not reveal any stable component of magnetisation and all directions, after heating to 300°C, changed but remained random. Repeat measurements after a few weeks showed large changes in direction so it is concluded that this sequence is unsuitable for palaeomagnetic study and is not considered further.

(ii) *Lower and Middle Old Red Sandstone* were sampled in the red facies bordering the Orcadian Basin. There were 17 sites (19–28, 32–34, 39–43) in the Middle Old Red and 10 sites (29–31, 35–38, 44–46) in the Lower sequence. Both showed low intensity of magnetisation and susceptibility, 10^{-6} gauss and 10^{-6} gauss oersted^{-1}. Initial directions of remanence within individual sites were somewhat scattered, with only 8 Middle and 2 Lower Old Red sites having directions grouped within a circular standard deviation of 15° or less (Table 1 and Fig. 2). After heating to 300°C, the within-site scatter was still large in most sites and only 7 Middle and 3 Lower Old Red sites contained closely grouped sample directions (Table 1 and Fig. 2). Pilot samples showed variable behaviour with most of them indicating consistent directions of magnetisation between 100 and 400°C, but these directions, even when well defined, were scattered from one site to the next (Table 2 and Fig. 3) although there was a distinct grouping of directions around 190°, −35°. This direction also seemed to be a favoured one for the more closely grouped site directions both before and after heating to 300°C (Fig. 2). After heating to 600° and higher, the sample directions showed no systematic pattern indicating that any stable remanence had already been demagnetised.

The behaviour of pilot specimens indicates that there is a large variation in stability within these Devonian sediments and, as some stable pilot specimens are derived from

TABLE 1

Sites Mean Directions with a circular standard deviation of 15° or less.

Mean directions calculated giving equal weight to each sample. The overall mean, after thermal demagnetisation, is 190°, −39°, giving equal weight to each site, with k = 11, alpha = 15°. Selecting sites with alpha less than 10°, the corresponding mean direction is 188°, −34·9°, k = 33, alpha = 13°.

Site Number	No. Samples	No. Specs.	Initial Decl.	Initial Incl.	300°C. Decl.	300°C. Incl.	alpha	c.s.d.
Middle Old Red Sandstone								
19	3	9	191	−36	190	−45	8	5
20	3	9	189	−39	191	−49	6	4
21	2	6	185	−35	184	−39	34	11
23	3	9	174	+15	192	−28	9	6
27	2	6	225	−40	217	−51	34	11
28	3	9	—	—	210	−40	22	14
32	4	4	272	+61	—	—	—	—
39	3	9	180	−33	181	−37	13	9
Lower Old Red Sandstone								
29	2	6	180	+10	187	−15	6	2
36	4	4	20	+56	—	—	—	—
38	6	6	—	—	203	−77	10	12
42	2	2	—	—	167	+5	45	14

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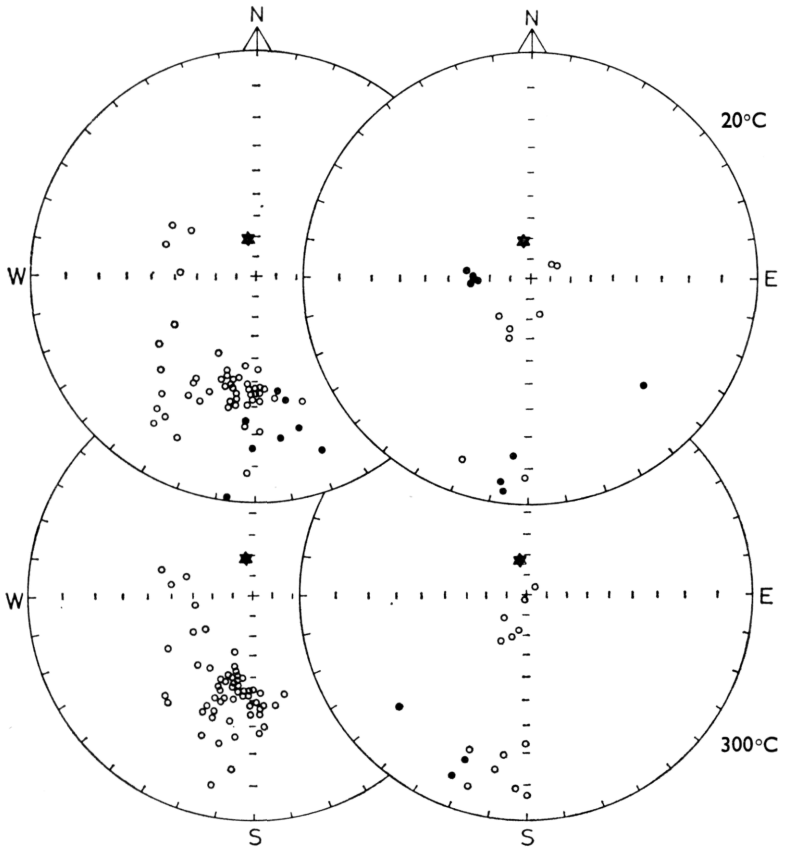


FIG. 2(a). Sample directions, before and after demagnetisation at 300°C. (a) Directions for samples from sites in which the internal scatter (circular standard deviation) is less than 15°. (b) Directions for samples from sites with higher internal scatter. The stereograms are equal angle projections, with the circumference as horizontal. Downward (positive) inclinations are shown as solid dots and upward inclinations as circles. All directions have been corrected for the tilt of the bedding. The direction of the present geomagnetic field (starred) is shown without correction for tectonic tilt as this varies from site to site, but as these corrections are small, its direction would move by only a few degrees so that the point illustrated can be taken as the generalised present geomagnetic direction.

sites with a large scatter, this stability appears to be inhomogeneously scattered both within and between sites so that only a few sites yield closely grouped directions of remanence. This uneven distribution of stability suggests that different ages of magnetisation are likely to be represented but it is significant that there is no clear grouping towards the present geomagnetic field direction. Only one direction of magnetisation, approximately 190°, −35°, appears to be indicated consistently by both the stable pilot specimens and sites with low internal scatter. Comparison of the pole positions (Fig. 4) corresponding to these more stable directions with those now available for Europe from Silurian to Tertiary times indicates that the dominant stable component in

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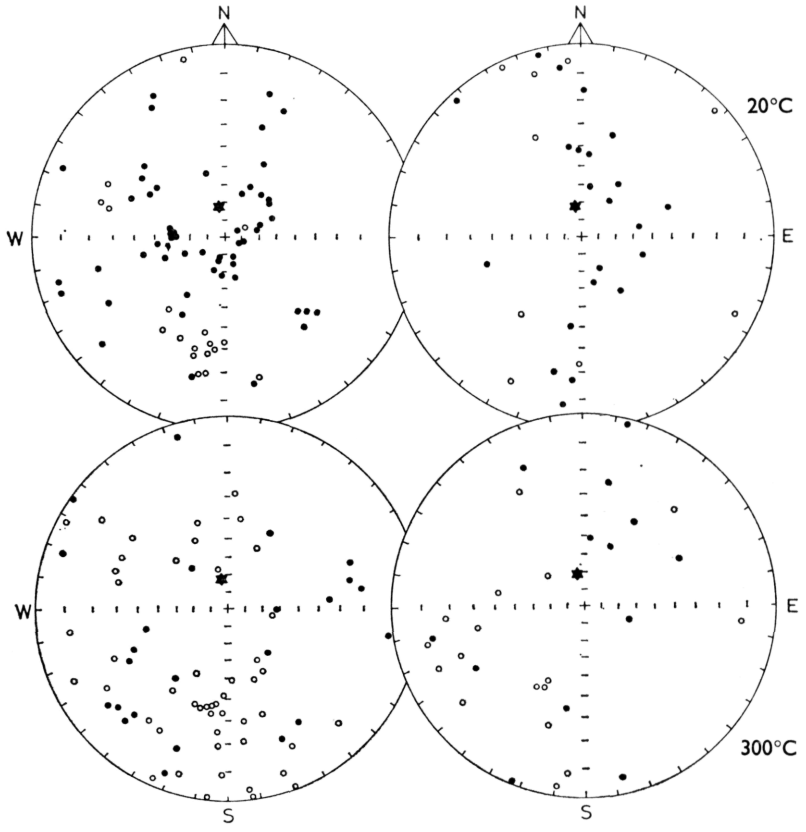


FIG. 2.(b)

these sediments were acquired in Carboniferous to Permian times, although some individual samples indicate ages of magnetisation which may well be Devonian. As the stable directions are all reversed, then the most likely age is during the Kiaman reversed period from late Westphalian until late Permian or Scythian. There is no indication of a magnetisation having been acquired in Triassic or later times. On this basis, it appears that many of the Devonian sediments in the Orcadian Basin contain haematite that was formed mainly in Permo-Carboniferous times and that this obscured or replaced the magnetisation of pre-existing iron oxides.

Sedimentological Aspects

The Old Red Sandstone of the Orcadian Basin has long been recognised as a continental facies of the Devonian (Crampton and Carruthers 1914). The Upper Old Red Sandstone consists almost entirely of fluvial deposits while the Middle, and to a lesser extent the Lower, Old Red Sandstone comprise rhythmic alternations between alluvial and lacustrine lithologies (Crampton and Carruthers 1914; Donovan *et al.* 1974; Collins and Donovan 1975) attributable to changes in water levels of the Orcadian lake.

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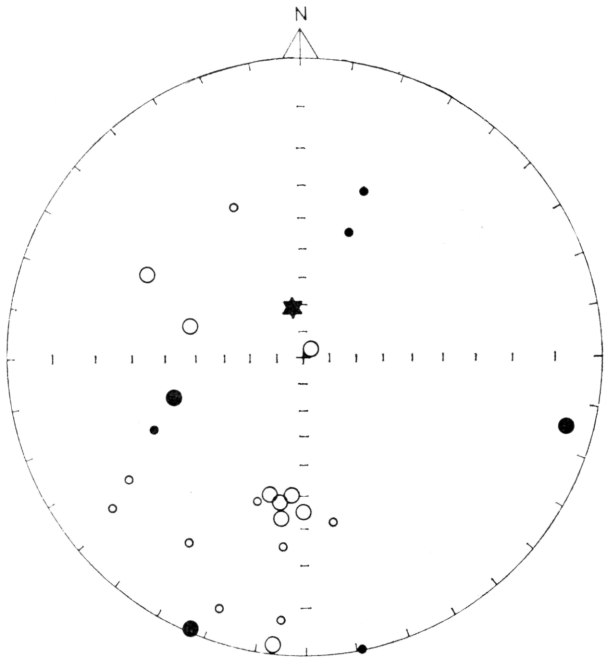


FIG. 3. Most stable directions of remanence isolated in pilot specimens during partial thermal demagnetisation. Projection and symbols as in Figure 2. Larger symbols correspond to directions with a stability index (Tarling & Symons 1967) greater than 5·0 (very high stability).

The samples in this study were taken from red horizons in the alluvial facies. The red colour (5R and 10R shades of Goddard, 1951) is due almost entirely to haematite occurring as particulate coatings around clastic grains and, in the more poorly sorted sediments, as well-shaped hexagonal crystals up to 0·7 μm diameter within the argillaceous matrix (Wilson 1971). Detrital haematite is present in small amounts in all the red beds, but in some areas, considerable accumulations have formed heavy mineral bands in which there is evidence of authigenic modification similar to that noted by Turner (1974) in the Ringerike Group of Norway. Haematite is also present as phyllosilicate pseudomorphs after micas, probably detrital biotite and chlorite, as some micas, particularly chlorite, are not completely oxidised and their textures resemble those of detrital micas in the Ringerike Sandstone. In the drab, often grey-black, lacustrine facies, the chlorite and biotite are fresh and clearly of detrital origin, although in these organic-rich beds, they are subordinate in quantity to muscovite.

The original environment of deposition was almost certainly oxygenated and conducive to the deposition of hydrous ferric oxides which, if the Eh-pH of the interstitial environment remained similar, would be expected to recrystallise to haematite at an early date, that is within the Devonian. This expectation is supported by (1) the intimate relation of red coloration to facies and the sharp definition of the facies boundaries; (2) mud flakes in intra-formational mud-flake breccias retain their colour, irrespective

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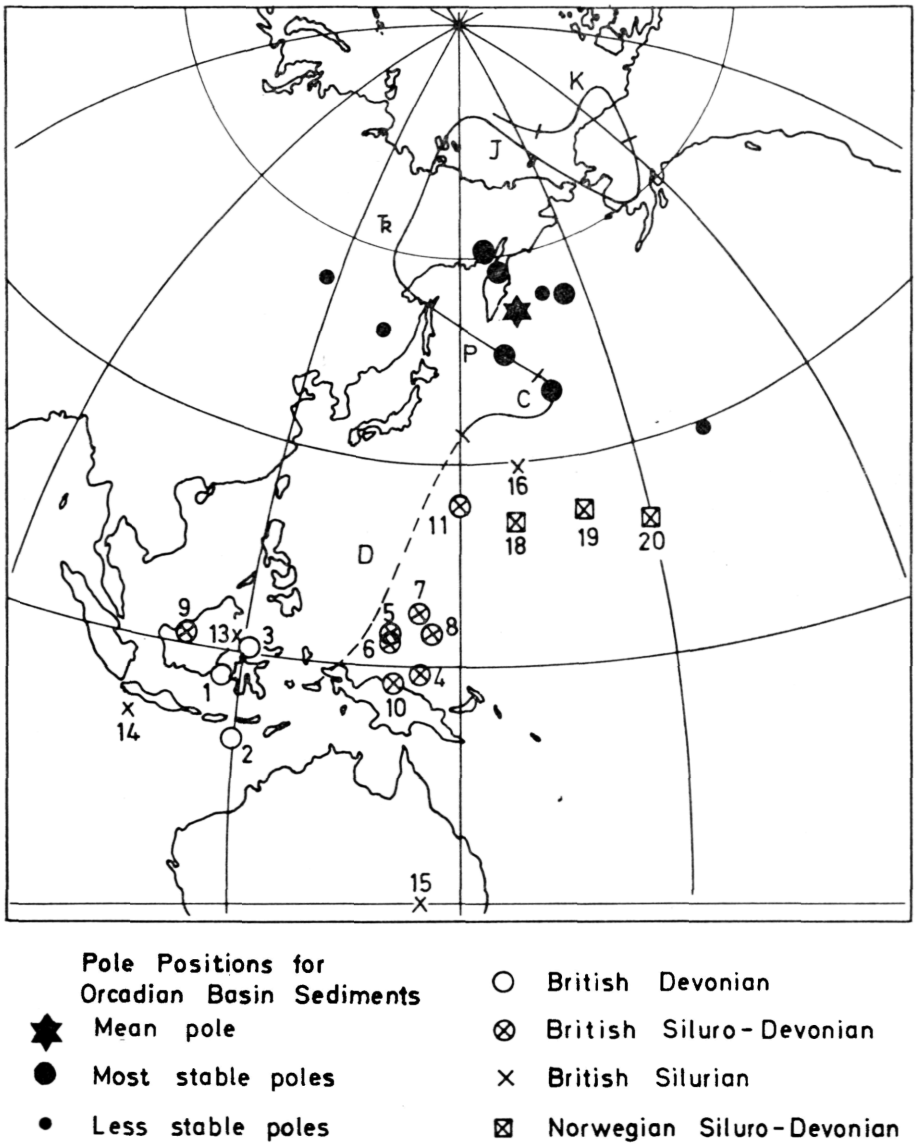


FIG. 4. European Siluro-Devonian pole positions and the Late Palaeozoic-Mesozoic polar wandering path. Pole Path-K = Cretaceous; J = Jurassic; P = Permian, C = Carboniferous; Supplementary Publication No. SUP 24002 deposited with the British Library, Boston Spa, contains information and references relating to poles 1–20.

of the colour of the host rock; (3) a small percentage of clastic grains have much thicker haematite coatings than usual (these may well have been recycled from a penecontemporaneously reddened sediment); (4) the frequent presence of kaolinite (Wilson 1971; Foster 1975) in the alluvial facies. This clay requires an open, acid pH environment for its formation. In the lacustrine facies, the clay mineral assemblages are dominated

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by illite and chlorite and do not contain kaolinite. It is possible to construct a generalised sequence of diagenetic events in these sediments, although not all stages are present in any one rock and the precise sequence of events is not always clear:—

1. Formation of particulate grain coatings of ferric oxide.

2. Pressure solution and the development of quartz overgrowths (this sometimes pre-dates stage 1).

3. Cementation by ferroan calcite, probably under low Eh conditions.

4. Dolomitisation of the carbonate cement commonly occurred.

5. Diagenetic growth of illite and chlorite to form overgrowths on detrital micas in some red beds, particularly the older ones. This stage appears to post date stages 1 and 2, but its relationship with stages 3 and 4 is not clear.

6. Ferric oxide formation by weathering of the ferroan carbonate cement leading to an increase in porosity. In many of the red beds the amount of cement is too low for this process to contribute significantly to the coloration and the finer-grained beds, in particular, contain little cement and have effective surface porosities of less than 1%.

This synopsis of the diagenetic history of these sediments clearly indicates an early date for haematite formation, certainly in the Devonian, with the minor exception of weathering of the cement (stage 6), which implies that the magnetisation carried by the haematite should be also of Devonian age.

Interpretation and conclusions

The palaeomagnetic studies indicate that the magnetisation of these sediments was acquired over a range of times from Devonian until possibly as late as Triassic, but with little or no later magnetisation and most of the magnetisation having been acquired during the Permian period. The sedimentological evidence clearly indicates a primary oxidising environment and the expectation that the crystallisation of haematite occurred during Devonian times and generally very shortly after deposition. There appear to be two possible explanations of this apparent contradiction:

1. The initial haematite formation could have been delayed until mainly Permian times. This could have been due to a radical change of the Eh-pH interstitial environment by, for example, the percolation of reducing groundwaters from the organic-rich lacustrine facies. Similarly, the natural maturation of the detrital chlorites and biotites (Walker 1967a, b), for example, may have been delayed for some 100 million years by the initial environmental conditions.

2. Haematite recrystallisation during some Permian event. This event is unlikely to be burial, or unroofing, as the low degree of compaction of the Dunnet Sandstones and the presence of very small quantities of montmorillomite in other localities suggests that these sediments have never been deeply buried (Weaver and Beck 1971). The more likely possibility is that haematite growth was re-activated by the deep penetration of oxygen during the desert weathering conditions of the Permain and further growth was stopped when the predominantly erosional conditions became predominantly

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depositional during the Triassic. Secondary weathering during the Permian is widespread and has been noted previously in British and Norwegian rocks of Siluro-Devonian age in the Anglo-Welsh Cuvette (Chamalaun and Creer 1964), the southwest of England (Cornwell 1966), the Fen Complex (Storetvedt 1973) and the Skomer Volanic Group (Briden *et al.* 1973). In the Midland Valley of Scotland, (Embleton 1968), secondary weathering of this period is known to have penetrated at least 650 m (Mykura 1960).

Nonetheless, although Permian weathering appears to be the most likely cause for the oxidation of the area, it is still obscure how this oxidation affected the haematite already present or which forms of haematite now carry this remanence. It is proposed to continue these studies on the red sediments and to extend investigations to the drab lacustrine sediments, limestones and siltstones. At this stage, however, it is interesting to note how palaeomagnetic techniques can be applied to the dating of diagnetic processes involving the formation of haematite (or magnetite) and the relation of such processes to other factors, such as rates of organic decomposition, that may be of economic importance. In this example, the former explanation would suggest that katagenesis of the organic matter in the Devonian lacustrine sediments was completed by Permian times. In this specific area, these observations already cast serious doubts on the suitability of these sediments for more routine palaeomagnetic studies, particularly when such results are used for determining tectonic displacements (Waage and Storetvedt 1973; Storetvedt 1974). It also suggests that other anomalous palaeomagnetic data, such a that from the Orkney Islands (Storetvedt & Petersen 1972; Morris *et al.* 1973) may be attributable to similar complex histories of magnetisation.

Acknowledgements

We are very grateful to Drs P. Turner, R. Foster and R. Archer for their comments and advice, particularly as they are currently investigating allied problems in these sediments. This work formed the partial basis for MSc dissertations in the Department of Geophysics & Planetary Physics, University of Newcastle upon Tyne, and these authors (J. M. Abou-Deeb and S. I. El-Batrouk) particularly thank other members of staff of that department for their assistance.

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MS accepted for publication 19th July 1975